Course Name: Mobile Applications Development
Course Code: CSC 3109
Level of Course: Year 3: Semester 1
Credit Units: 4

Contact Hours:

- Lectures: 30 Hours
- Practical Sessions / Tutorials: 15 Hours
- **Total:** 45 Hours

Pre-requisites:

- Object-Oriented Programming (Java, Database Management Systems)

Course Description
In an era defined by ubiquitous connectivity, mobile computing has transcended its role as a mere communication tool to become the primary engine for global digital transformation. This course serves as a rigorous bridge between theoretical software engineering and the high-stakes reality of the global app economy. In this course, students will be immersed in the full software development life cycle (SDLC), in native Android (Kotlin) and cross-platform Flutter (Dart).

The course is strategically positioned within a robust Institutional and Professional Ecosystem designed to transition learners from academic settings to industry leadership. The course aligns with the standards of the IEEE Computer Society and the Google Developers Group (GDG), ensuring that architectural patterns like Clean Architecture meet global benchmarks. Nationally, the course integrates the regulatory and innovation frameworks of NITA-U and the Ministry of ICT & National Guidance, emphasizing compliance with Uganda's Data Protection Act and fostering participation in the National Innovations. The course leverages the Mbarara innovation corridor, collaborating with local hubs to provide students with real-world mentorship, incubation opportunities, and a community of practice. Ultimately, this course is not merely a technical module but a professional gateway, empowering students to deploy secure, scalable, and socially-impactful mobile solutions within the regional and global digital landscape.

Course Rationale

As digital services in health, finance, and governance move mobile-first, the demand for specialized developers is critical. This course equips students with the technical agility and professional ethical grounding required to navigate the resource-constrained and rapidly evolving environment of mobile platforms.

Course Aim and Objectives

a) **Course Aim**

To develop advanced conceptual, technical, and professional competence in the principles and practices of modern mobile software engineering within a collaborative industry ecosystem.

b) Specific Objectives

By the end of the course, learners will be able to:

- Analyze the architectural ecosystems of modern mobile platforms (Android/iOS).
- Design responsive and accessible mobile interfaces using industry-standard UI/UX tools.
- Implement secure data persistence and cloud synchronization mechanisms.
- Integrate third-party APIs and background services into full-stack mobile solutions.
- Deploy and monitor production-ready applications on global platforms.

Course Intended Learning Outcomes (CLOs)

- CLO1 (Knowledge): Explain mobile architecture patterns, application lifecycles, and security protocols.
- CLO2 (Application): Build responsive multi-screen UIs and manage local/cloud data persistence.
- CLO3 (Analysis): Evaluate mobile-specific constraints like battery life, memory usage, and network latency.
- CLO4 (Professional Practice): Execute a full deployment pipeline, including version control and app store release.
- CLO5 (Transfer): Apply mobile development skills to solve real-world problems through a functional prototype.

Competencies To Be Developed

a) Technical & Domain Competences

- Mobile systems architecture (MVVM/Clean Architecture)
- Native (Kotlin) or Cross-platform (Dart/Flutter) development
- Local (Room/SQLite) and Cloud (Firebase) data persistence

b) Analytical & Cognitive Competences

- Systems thinking for resource-constrained environments
- Critical evaluation of mobile UI/UX and accessibility

c) Professional, Ethical & Governance Competences

- Secure coding and compliance with Uganda’s Data Protection Act
- Version control (Git) and collaborative engineering workflows

d) Transferable & Leadership Competences

- Problem identification and requirement elicitation
- Project management and milestone tracking

Competence-Based Course Content Outline

Week	CLO	Key Content	Teaching & Learning Activities	Competences Emphasised	Assessment Tasks	Workload (Hrs)
1	CLO 1	Ecosystems & Ideation	Lecture & Brainstorming	Problem Identification	Concept Note	4
2	CLO 1	Architecture & Setup	Lab: Setup & Repo creation	Technical Readiness	Skeleton App	4
3	CLO 2	UI/UX & Wireframing	Design Studio	UI/UX Design	UI Prototype	4
4	CLO 2	Navigation & State	Coding Workshop	Logic & Flow	Navigation Map	4
5	CLO 2	Local Data (Room)	Practical Lab session	Data Modeling	CRUD Demo	4
6	CLO 2	Cloud Sync (Firebase)	Live Demo	Cloud Integration	Auth Module	4
7	CLO 3	Networking (APIs)	API Integration Lab	Systems Integration	JSON Parsing Task	4

8	CLO 3	Mid-Semester Defense	Project Presentation	Communication	Milestone Demo	5
9	CLO 2	Advanced State (MVVM)	Advanced coding lab	Architectural Integrity	State Mgmt Task	4
10	CLO 4	Background Tasks	Code Review / Studio	Performance Tuning	Background Svc	4
11	CLO 4	Testing & QA	Peer Testing Studio	Quality Assurance	Bug Log	4
12	CLO 4	Play Store Deployment	Publishing Workshop	DevOps/Deployment	Signed Bundle	4
13	CLO 5	Final Polish	Studio Feedback session	Attention to Detail	Final App Beta	4
14	CLO 5	Technical Documentation	Writing Workshop	Technical Writing	Final Report	4

Assessment Plan

Assessment Philosophy:

Assessment is project-driven, focusing on the incremental development of a functional product and the ability to defend architectural choices.

a**) Weighting by Learning Dimension**

Dimension	Weight	Description
Advanced Knowledge	20%	Understanding mobile architecture patterns (MVVM), lifecycles, and security protocols.
Application	30%	Building functional UIs, logic, and database integrations.
Analysis & Judgement	25%	Evaluating performance, battery efficiency, and cross-platform trade-offs.
Transfer & Professional Insight	25%	Producing technical reports, app store deployment, and user-centric problem solving.

b) Assessment Plan

Component	Mode of Assessment	Week	Weight
Initial Proposal	Problem statement, concept note, and low-fidelity wireframes.	3	10%
Project Prototype	Functional UI with navigation and local database integration.	7	15%
Mid-Semester Evaluation	Live demonstration of core features (Cloud/Auth/Logic).	9	15%

Practical Assignments	Coding labs on API consumption and background services.	11	20%
Final Examination	Functional App (25%) + Technical Report (10%) + Deployment (5%)	13-15	40%
Total			100%

Learning Resources

Core Textbooks:

1. Griffiths, D. (2021). *Head First Android Development* (3rd ed.). O'Reilly.
2. Windmill, E. (2020). *Flutter in Action*. Manning Publications.

Free Certifications:

- Google: Android Basics in Kotlin; https://developer.android.com/courses/android-basics-kotlin/course

- Meta: Android Developer Professional Certificate.; https://www.coursera.org/professional-certificates/meta-android-developer